

PROJECT-Q Workbook - 09

30-day Quantum Computing Challenge

DAY 09

Multi-Qubit Systems

Learning Objectives

By the end of this chapter, you will be able to:

- ✓ Understand why multiple qubits are needed.
 - ✓ Represent two-qubit and three-qubit quantum states.
 - ✓ Learn the tensor product in an intuitive way.
 - ✓ Understand computational basis states.
 - ✓ Build simple multi-qubit circuits in Qiskit.
 - ✓ Calculate the dimension of multi-qubit systems.
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Quick Recap

Previously, you learned:

- A single qubit can exist in a superposition.
- Quantum gates manipulate qubits.
- Quantum circuits describe quantum computations.
- Qiskit provides the software framework for building and executing quantum programs.

Today, we move from **one qubit** to **multiple qubits**, unlocking exponentially larger state spaces.

Why Do We Need More Than One Qubit?

Imagine trying to write a paragraph using only one letter. No matter how many times you reuse that letter, you cannot express meaningful information.

Similarly, a single qubit is useful for learning, but practical quantum algorithms require many qubits working together.

Examples include:

- Quantum search
- Quantum cryptography
- Error correction
- Optimization
- Drug discovery

The real power of quantum computing begins when qubits interact.

Revisiting a Single Qubit

A single qubit is written as

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

where

$$|\alpha|^2 + |\beta|^2 = 1$$

This tells us that the total probability of measuring the qubit must always equal 1.

From One Qubit to Two Qubits

A common misconception is:

"Two qubits simply give four states."

Not exactly.

Two qubits form **one combined quantum system** whose state is described using the **tensor product**.

Instead of writing two independent qubits, we combine them into a single state space.

The Tensor Product (Intuition First)

Think of each qubit as adding a new "dimension" of information. When two qubits are combined, we don't add their possibilities, we multiply them.

For example:

One qubit:

0
1

Another qubit:

0
1

Together, the possible basis states become:

00
01
10
11

Notice:

$2 \times 2 = 4$ possible basis states.

This multiplication process is called the **tensor product**.

Mathematical Representation

The tensor product is represented using the symbol, $\otimes$

For example,

$$|0\rangle \otimes |1\rangle = |01\rangle$$

Similarly,

$$|1\rangle \otimes |0\rangle = |10\rangle$$

In practice, the $\otimes$ symbol is usually omitted once the combined state is written.

Computational Basis States

For two qubits, there are four computational basis states:

$$|00\rangle, |01\rangle, |10\rangle, |11\rangle$$

Each basis state represents one definite outcome when both qubits are measured.

State Vector of Two Qubits

Unlike a single qubit, a two-qubit system requires four probability amplitudes.

The general state is,

$$|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$$

where,

$$|\alpha|^2 + |\beta|^2 + |\gamma|^2 + |\delta|^2 = 1$$

This is simply the same normalization rule extended to four basis states.

How Does the State Space Grow?

One of the most remarkable properties of quantum computing is **exponential growth**.

Number of Qubits	Basis States
1	2
2	4
3	8
4	16
5	32
n	(2^n)

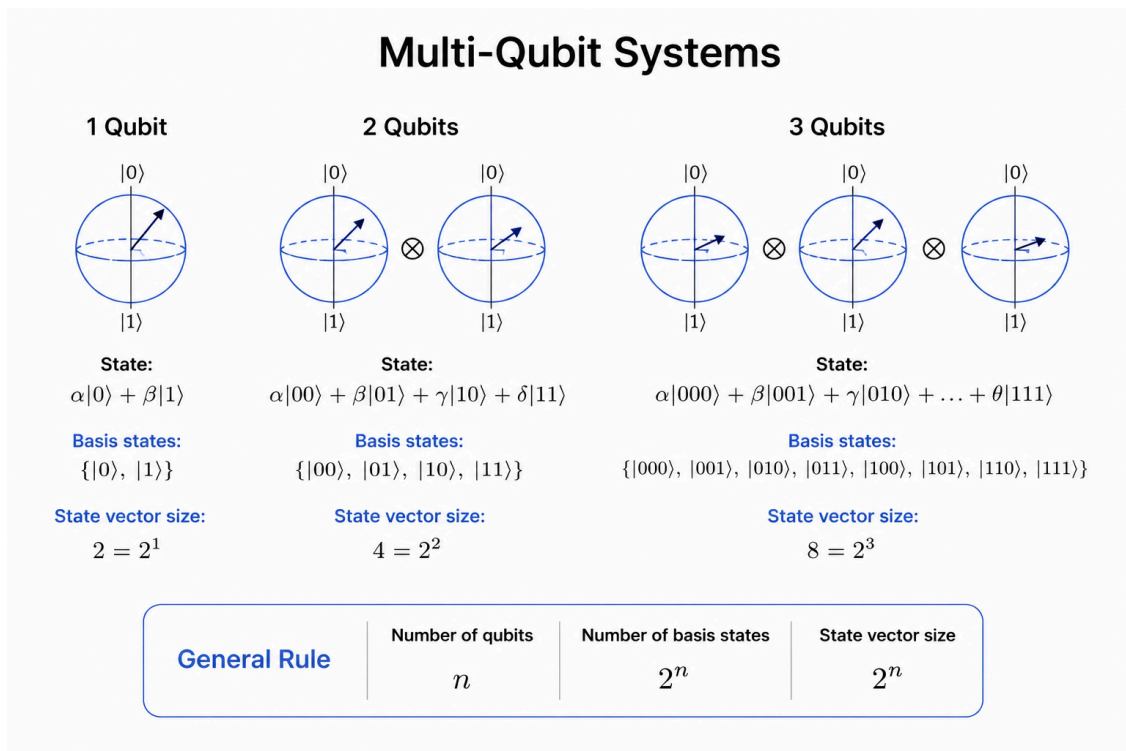
Every additional qubit doubles the size of the state space.

This exponential scaling is one of the reasons quantum computers can outperform classical computers for certain problems.

Visualizing Two-Qubit States

Instead of a Bloch Sphere, it is often helpful to imagine a table of possible outcomes.

		Second	Qubit
		0	1
First	0	00	01
Qubit	1	10	11

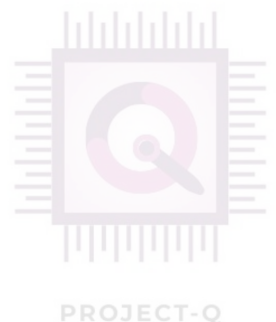


Each cell represents one computational basis state.

Multi-Qubit Circuits in Qiskit

Creating multiple qubits is straightforward.

```
from qiskit import QuantumCircuit
qc = QuantumCircuit(2,2)
```



This creates:

- 2 qubits
- 2 classical bits

You can apply different gates to each qubit independently.

```
qc.h(0)
```

```
qc.x(1)
```

Here,

- the first qubit enters superposition,
- the second qubit undergoes a bit flip.

Measuring Multiple Qubits

Measurement is performed independently for each qubit.

```
qc.measure([0,1],[0,1])
```

Possible outputs include:

00

01

10

11

The exact probabilities depend on the operations performed before measurement.

Worked Example

Suppose both qubits begin in $|00\rangle$

Apply

H on q0

x on q1

After these operations, the first qubit is in superposition while the second is fixed at $|1\rangle$.

When measured, the system can produce:

$|01\rangle$ or $|11\rangle$

each with approximately equal probability.

This demonstrates how each qubit contributes independently to the overall system when no entangling operations are applied.

Why This Matters

With multiple qubits, quantum computers can:

- Represent exponentially larger state spaces.
- Perform parallel quantum operations through superposition.
- Prepare for entanglement and controlled operations.
- Execute practical quantum algorithms.

This is the point where quantum computing begins to show capabilities beyond single-qubit demonstrations.

Common Beginner Mistakes

✗ Two qubits are just two separate single qubits.

✓ Once combined, they form a single quantum system described by a larger state vector.

✗ Two qubits store twice as much information.

✓ They provide a state space with **four basis states**, not just double the information.

✗ Tensor product means adding qubits.

✓ Tensor products **combine** quantum systems by multiplying their state spaces.

Remember This

- Two qubits have four computational basis states.
- Multi-qubit systems are represented using the tensor product.

- The state space grows exponentially as (2^n) .
- The normalization rule still applies: total probability equals 1.
- Multiple qubits are the foundation for controlled gates and entanglement.

Math Corner:

FORMULA SHEET

Multi-Qubit Systems

Concept	Formula	Explanation
1. Single-Qubit State	$ \psi\rangle = \alpha 0\rangle + \beta 1\rangle$	General state of a single qubit, where α and β are complex probability amplitudes.
2. Probability of Measurement	$P(0) = \alpha ^2$ $P(1) = \beta ^2$	Probability of measuring 0 or 1 is the squared magnitude of their amplitudes.
3. Normalization (1 Qubit)	$ \alpha ^2 + \beta ^2 = 1$	The total probability must always equal 1.
4. Tensor Product	$ a\rangle \otimes b\rangle = ab\rangle$	Combines two quantum states into a single composite state.
5. Computational Basis (2 Qubits)	$ 00\rangle, 01\rangle, 10\rangle, 11\rangle$	The four basis states of a two-qubit system.
6. Two-Qubit State	$ \psi\rangle = \alpha 00\rangle + \beta 01\rangle + \gamma 10\rangle + \delta 11\rangle$	General state of two qubits with four probability amplitudes.
7. Normalization (2 Qubits)	$ \alpha ^2 + \beta ^2 + \gamma ^2 + \delta ^2 = 1$	The sum of probabilities of all basis states must equal 1.
8. Number of Basis States (n Qubits)	2^n	A system of n qubits has 2^n computational basis states.
9. Dimension of State Vector (n Qubits)	2^n	The state vector of an n -qubit system has 2^n complex amplitudes.
10. General n-Qubit State	$ \psi\rangle = \sum_{x=0}^{2^n-1} \alpha_x x\rangle$	General state of an n -qubit system, where $ x\rangle$ represents each basis state.
11. Normalization (n Qubits)	$\sum_{x=0}^{2^n-1} \alpha_x ^2 = 1$	The sum of probabilities of all 2^n basis states must equal 1.
12. Example State	$ \psi\rangle = \frac{ 00\rangle + 11\rangle}{\sqrt{2}}$	Probability of $ 00\rangle = 1/2$, Probability of $ 11\rangle = 1/2$, Others = 0.

End-of-Day Summary

Today, you learned:

- Why multi-qubit systems are necessary.
 - How multiple qubits are represented mathematically.
 - The concept of the tensor product.
 - Computational basis states.
 - Two-qubit state vectors.
 - Exponential growth of quantum state spaces.
 - Building and measuring multi-qubit circuits in Qiskit.
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Top Verified Learning Resources

1. **Medium - Basic Quantum Computing: Multi-Qubit Systems** (*Good conceptual explanation with mathematical intuition*)
https://medium.com/@charlie.thomas_94667/basic-quantum-computing-multi-qubit-systems-4f1851a831f3
2. **Medium - Learn Quantum Computing with Qiskit: Multiple Qubits** (*Useful for Qiskit-based examples and circuit implementation*)
https://medium.com/@_monitsharma/learn-quantum-computing-with-qiskit-multiple-qubits-and-entanglement-39b8ffa9b95d
3. **Video - Multi-Qubit Systems (Recommended)**
<https://www.youtube.com/watch?v=49ViWT1--Z4>